

Dear Editor,

We are writing to enquire whether our manuscript, “NestNetR: Deep-learning based analysis of breeding behaviour from light and temperature recordings in R”, would be suitable for consideration in *Ornithological Applications*.

Biologging is rapidly expanding our ability to study avian behaviour, but the pace and volume of time-series data now exceed what can be processed consistently with manual or ad hoc approaches. As a result, there is an increasing need for objective, scalable, and reproducible workflows that convert biollogger recordings into comparable behavioural metrics, particularly in breeding ecology, where robust inference on incubation, brooding, and breeding phenology underpins population-level analyses and conservation decisions, especially in remote regions with limited field access.

In this study, we introduce **NestNetR**, an open-source R package that provides a **fully integrated workflow** for behavioural classification from light-level geolocator data using deep learning. NestNetR supports data import and preprocessing, model use or custom model definition, and automated **daily classification of breeding-related behaviour** within a modular, reproducible framework implemented entirely in R. Using an extensive Arctic breeding-season dataset of shorebirds, we demonstrate that breeding behaviours relevant to reproductive success can be inferred accurately from combined light and temperature time series, and that trained models can be transferred across species.

We believe the manuscript aligns with the scope of *Ornithological Applications* by providing a practical and transferable analytical tool that enables the extraction of biologically meaningful breeding metrics from widely used avian biologging devices. These outputs have direct relevance for studies of breeding phenology, reproductive investment, and responses of bird populations to environmental change, with clear applications to conservation research and management.

We would appreciate your guidance on whether this manuscript fits the scope of *Ornithological Applications*.

Thank you very much for your time and consideration.

Kind regards,

Simeon Lisovski

(on behalf of all authors)